

Subject-Disjoint Benchmarking of Lightweight Detectors for Visible Driver-Cue Localization

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Abstract—Road traffic injuries remain a leading cause of death worldwide, and driver monitoring systems (DMSs) are increasingly mandated by vehicle-safety assessment programmes. Object detection can spatially localize visible evidence of driver inattention inside a vehicle cabin, yet existing evaluations often lack subject-disjoint splits, retain no negative frames, or select operating points on test data. We present a reproducible benchmark that addresses these gaps. The primary red–green–blue (RGB) experiment retains 15,723 Driver Monitoring Dataset frames from 14 drivers, including 12,722 negatives, and evaluates three nano-scale detection systems — You Only Look Once (YOLO) 11n, YOLO26n, and nano Fine-grained Distribution Refinement (D-FINE-N) — across three paired seeds under a strict validation-only threshold protocol. No single detector dominates: YOLO11n leads coarse-IoU mean average precision (mAP₅₀) at $91.79 \pm 0.44\%$ and throughput at 68.0 ± 1.3 frames per second (FPS), while D-FINE-N leads micro- F_1 at $91.14 \pm 2.66\%$ but runs slowest at 40.4 ± 1.2 FPS. Between-subject variation exceeds seed variation, and class imbalance materially affects pooled metrics. A secondary near-infrared (NIR) training-negative exposure study remains [PLACEHOLDER]. Claims concern visible-cue localization, not temporal fatigue or behavior inference.

Index Terms—driver monitoring, object detection, subject-disjoint evaluation, visible driver cues, near-infrared imaging, real-time inference

I. INTRODUCTION

Road traffic injuries claim approximately 1.19 million lives annually and impose a substantial economic burden worldwide [1]. In response, Euro NCAP now awards safety-assist credit for camera-based driver monitoring [2], accelerating the deployment of in-cabin driver monitoring systems (DMSs). These systems must operate robustly under changes in driver identity, illumination, viewpoint, and occlusion. Whole-frame recognition can describe an activity, but it may also exploit identity or cabin context without showing the visual evidence used for a decision. Object detection provides a complementary formulation: it localizes cue-specific evidence, makes false positives and misses spatially interpretable, and measures detections on frames containing no target cue. A single-frame detector also avoids the latency and architectural complexity of a temporal model when real-time throughput is the experimental focus. This formulation does *not* infer fatigue, intent, duration, or another temporal driver state.

DMD provides multimodal attention and alertness recordings [3], while Drive&Act contains synchronized color, NIR, depth, and pose streams for fine-grained activities [4]. Both contain

temporally correlated video and repeated observations of each driver. Evaluation can therefore be optimistic if neighboring frames or identities cross partitions, natural negative frames are removed, the test set determines the confidence threshold, or latency boundaries differ. Existing driver-detection studies on DMD [5] and posture-classification benchmarks such as the State Farm challenge [6] further illustrate the diversity of evaluation choices and the difficulty of cross-paper comparison.

Recent work proposes specialized lightweight networks for in-cabin recognition and detection [7]–[10]. Our objective is different: to compare complete, publicly released nano detector systems without task-specific architectural modification. The contributions are:

- a primary subject-disjoint RGB benchmark for four explicitly defined visible cues, with all sampled frames preserved and annotation structure automatically audited;
- validation-only operating-point selection followed by protected testing, three paired RGB seeds, class- and subject-level analysis, false detections on negative frames, and synchronized latency; and
- a checksum-backed reproduction workflow plus a clearly separated, one-seed, five-system NIR training-negative exposure study with deterministic 1-FPS midpoint sampling.

II. RELATED WORK

A. Driver Recognition and Cue Localization

MobileVGG and convolution–transformer hybrids demonstrate efficient distracted-driver classification [7], [11]. PO-GUISE+ uses pose and object guidance for efficient driver-action recognition in Drive&Act NIR imagery [12]. Such whole-frame or temporal recognition tasks are not interchangeable with the present bounding-box task. Here, a correct prediction must identify both a cue class and its spatial evidence in one frame; no temporal state is estimated.

Detection-specific work modifies YOLO backbones, attention, or multiscale aggregation for dangerous-driving or fatigue cues [8]–[10]. Comparative studies assess convolutional neural networks (CNNs) against transformers [13] or several YOLO generations [14]. The YOLO family originates with single-shot grid regression [15] and has evolved through multiple architectural revisions [16]–[18]. However, differences in driver separation, retained negatives, operating-point calibration, and runtime protocol limit direct cross-paper ranking.

TABLE I
FROZEN RGB BOX SEMANTICS AND TEST SUPPORT

Cue	Bounding-box target	Test n	Median area
Yawning	Mouth	33	1.38%
Hand over mouth	Full face and covering hand	28	24.67%
Drinking	Visible hand/container-mouth interaction	56	16.09%
Phone use	Visible hand-phone interaction	497	8.37%

B. Released Detector Systems

The Detection Transformer (DETR) casts object detection as bipartite set prediction [19]; the Real-Time Detection Transformer (RT-DETR) makes this formulation practical for real-time use [20]; and D-FINE refines box-edge distributions for improved localization [21]. EfficientDet instead compounds an EfficientNet backbone, bidirectional feature pyramid network (BiFPN), and anchor heads [22], while RTMDet is an anchor-free one-stage family built around CSPNeXt and a path-aggregation pyramid [23]. The RGB comparison uses D-FINE-N, YOLO11n [17], and YOLO26n [18] as shipped systems. The exploratory NIR comparison adds EfficientDet-D1 and RTMDet-Tiny to broaden the architecture families. Native training and inference recipes remain part of each system: the pinned YOLO26 and D-FINE paths are end-to-end, whereas YOLO11n, RTMDet-Tiny, and EfficientDet-D1 retain their required non-maximum suppression (NMS) [24]. Recent edge-deployment work demonstrates that such lightweight systems can approach real-time performance on low-cost hardware [25]. This is therefore a system-level benchmark, not an isolation of architecture alone.

III. METHODOLOGY

A. Task Definition and Workflow

The task is bounding-box localization of four visible RGB cues: yawning, hand over mouth, drinking, and phone use. Detection is used because it spatially localizes the evidence, discourages whole-frame identity and cabin shortcuts, and exposes errors on negative frames. Figure 1 summarizes the shared protocol. Dataset construction, splitting, training, and validation precede a checksum and confirmation gate; no test-derived threshold or model change is permitted. The RGB benchmark is primary. NIR is a distinct secondary experiment rather than a paired modality comparison.

B. RGB Dataset and Annotation Protocol

The RGB dataset contains 15,723 face crops sampled at 1 FPS from 81 DMD recordings of 14 drivers. One primary annotator labeled the data in a single smooth pass; no independent second-person review was established. Every sampled frame was preserved to avoid post-sampling curation, yielding 3,001 positive and 12,722 negative frames. Table I freezes the spatial ontology. Box semantics intentionally differ because the visible evidence differs by cue.

Each frame contains zero or one box. Yawning and hand-over-mouth are mutually exclusive; when a hand covers a

TABLE II
SHARED RGB TRAINING CONFIGURATION

Setting	Value
Input resolution	640 × 640
Physical / effective batch	8 / 32 (4-step accumulation)
Epochs	220 (no early stopping)
Precision	AMP (FP16/FP32)
Pretrained backbone	COCO
Seeds	{13, 37, 73} (predeclared)
YOLO optimizer	auto + linear schedule
D-FINE optimizer	AdamW + MultiStepLR

yawn, hand-over-mouth takes precedence. Boxes enclose only the visible target. Occluded, truncated, or out-of-frame anatomy is never estimated, although a partially occluded cue remains valid when visually identifiable. No arbitrary minimum-pixel cutoff is applied at 640 × 640.

An automated COCO audit found 15,723 unique images and 3,001 unique annotations, with zero duplicate identifiers or file names, orphan annotations, invalid categories, nonfinite or nonpositive boxes, out-of-bounds boxes, or multi-annotation frames. The smallest annotated area was 0.81% of the frame; 139 of 159 yawning boxes occupied below 2%. This is a structural audit, not evidence of inter-annotator agreement or semantic correctness.

C. Split, Models, and Training

An exhaustive 8:3:3 subject assignment minimizes the worst relative class-distribution deviation, with root-mean-square deviation as a tie-breaker. The frozen training, validation, and test partitions contain 9,087, 3,423, and 3,213 frames. The three test subjects contribute 1,080, 920, and 1,213 frames and 178, 183, and 253 positives, respectively.

COCO-pretrained YOLO11n [17], YOLO26n [18], and D-FINE-N [21] use 640 × 640 input, physical batch size 8, four-step gradient accumulation for an effective batch of 32, 220 epochs, automatic mixed precision (AMP), and no early stopping. A fixed patch normalizes an incomplete final accumulation window by its actual sample count. Native recipes are retained: Ultralytics optimizer=auto with a linear schedule, and the official D-FINE AdamW/MultiStepLR recipe. Seeds {13, 37, 73} were declared in advance, and no run was selected for presentation. Table II consolidates the shared settings.

D. Validation, Test Metrics, and Profiling

The best model-native validation checkpoint is retained. Its confidence threshold τ^* maximizes validation micro- F_1 over [0.01, 0.99], with higher precision and then higher τ as tie-breakers. At an intersection over union (IoU) of 0.5, predictions are greedily matched one-to-one within class. Protected testing reports COCO mAP₅₀ and mAP_{50:95} [26], precision, recall, micro- and macro- F_1 , and

$$FD_{100} = 100 \frac{FP_{neg}}{N_{neg}}, \quad (1)$$

where FP_{neg} is the number of above-threshold false-positive detections on negative frames. We introduce FD_{100} as a simple,

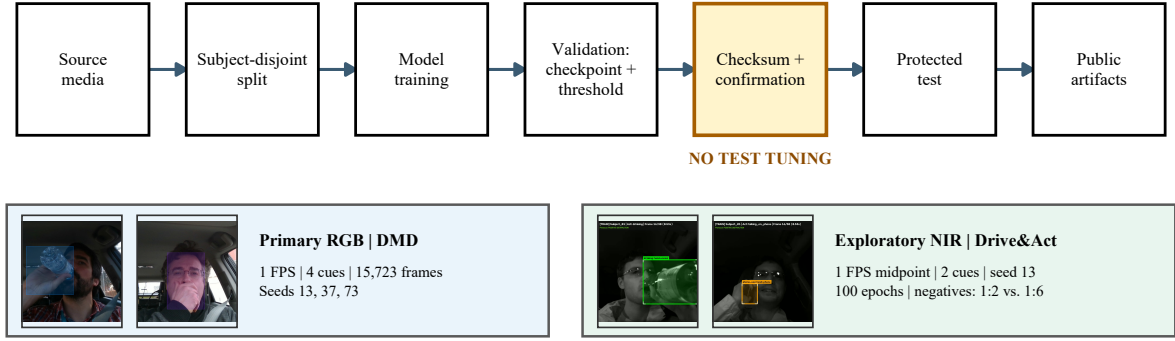


Fig. 1. End-to-end benchmark workflow with representative annotated DMD and Drive&Act frames. RGB and NIR share the validation-before-test gate but retain dataset-specific sampling, ontology, seeds, and training conditions. FPS denotes frames per second.

interpretable diagnostic: it counts how often the detector fires on frames where no target cue is present. It is a static per-frame count, not a modeled nuisance-alert rate; no temporal confirmation or debouncing is applied.

RGB results are the sample mean and sample standard deviation (SD) over three seeds. Because the subject split is fixed, SD measures optimization variability, not generalization uncertainty across held-out drivers. Each test subject is therefore reported separately, and paired per-seed differences are checked for sign consistency. No significance test is claimed for $n = 3$.

The six YOLO passes were completed before D-FINE-N was added. D-FINE-N selection remained validation-only and its three test manifests were frozen before evaluation; however, because the D-FINE runs were executed in a separate automation suite, the YOLO and D-FINE results are presented together only via the shared test partition, not via a single joint run artifact.

Batch-1 16-bit floating-point (FP16) profiling uses an NVIDIA GeForce RTX 4060 8-GB graphics processing unit (GPU). CUDA synchronization brackets model-forward and tensor-to-final-detection intervals. Disk input/output, decoding, preprocessing, and metric computation are excluded. We report p50 tensor-to-final latency and sustained FPS; these measurements do not represent central-processing-unit or embedded-device deployment.

E. Exploratory NIR Protocol

The secondary NIR source pool contains 3,786 one-second front-top active-NIR (850 nm) Drive&Act snippets from 15 drivers. Twenty-three unilluminated task snippets are excluded before splitting, leaving 3,763. Drinking and phone-use tracklets provide two positive classes. Each snippet contributes one deterministic frame at 1 FPS: the temporal midpoint at $t = 0.5$ s, or zero-based source-frame offset 14. The corresponding box is linearly interpolated at the same instant, then the driver crop is resized to 640×640 . Unlike RGB, NIR therefore has dataset-specific frame exclusion and interpolated annotations.

A fixed 9:3:3 subject partition is shared across all ten model-condition runs. The five systems span convolutional YOLO,

anchor-free CSPNeXt/PAFPN (RTMDet-Tiny), anchor-based EfficientNet-B1/BiFPN (EfficientDet-D1), and transformer (D-FINE-N) families. Validation and test contain identical ordered examples. Training always contains 270 positive snippets and either 540 or 1,620 negative snippets, corresponding to nominal positive:negative ratios of 1:2 and 1:6. The 1:2 negatives are a subject-stratified seed-13 subset of the 1:6 pool.

All runs use physical batch 8, four-step accumulation, a fixed 100-epoch maximum without early stopping, and the best model-native validation checkpoint. Native optimizers and schedules are retained. The D-FINE augmentation transition is placed at epoch 67; RTMDet switches from cached Mosaic/MixUp to its weak stage at epoch 93; and EfficientDet retains momentum optimization, cosine decay, warm-up, and exponential moving average. The 1:2 and 1:6 conditions produce 2,600 and 6,000 optimizer updates, respectively. Thus, 1:6 has three times the unique negatives and approximately 2.31 times the optimization updates. It is consequently a *training-negative exposure study*, not a causal class-ratio ablation under matched training signal.

All NIR jobs use seed 13, and operating metrics follow the RGB protocol. Since one frame represents each snippet, frame- and snippet-level decisions coincide; equivalent snippet metrics are not duplicated in the publication tables.

IV. RESULTS AND ANALYSIS

A. Aggregate RGB Results

Table III summarizes all nine RGB runs. YOLO11n leads mAP₅₀ and throughput, YOLO26n leads mAP_{50:95}, and D-FINE-N leads micro- and macro- F_1 . The D-FINE advantage is concentrated at the calibrated operating point rather than strict-IoU AP and costs 8.0–26.4 FPS relative to the YOLO systems. The evidence supports an accuracy–efficiency trade-off, not universal superiority.

B. Class Imbalance and Paired-Seed Evidence

Phone use accounts for 497 of 614 positive test instances, whereas hand over mouth, yawning, and drinking have only 28, 33, and 56 instances. This imbalance can influence learned

TABLE III
MEAN AND SAMPLE STANDARD DEVIATION ACROSS THREE SEEDS FOR THE RGB TRACK ON THE PROTECTED TEST SPLIT. BOLD INDICATES THE BEST RESULT PER COLUMN.

Model	mAP ₅₀	mAP _{50:95}	Micro- F_1	Macro- F_1	FD ₁₀₀	p50 (ms)	FPS	Params (M)	GFLOPs
YOLO11n	91.79±0.44	55.38±0.41	86.60±0.49	83.42±1.36	0.63±0.04	14.30±0.20	68.0±1.3	2.59	6.44
YOLO26n	89.28±0.99	56.98±0.81	84.44±1.14	79.39±1.49	0.67±0.06	19.21±0.09	51.7±0.2	2.51	5.78
D-FINE-N	90.22±2.29	55.38±0.89	91.14±2.66	84.33±4.51	0.87±0.21	23.31±0.46	40.4±1.2	3.72	7.44

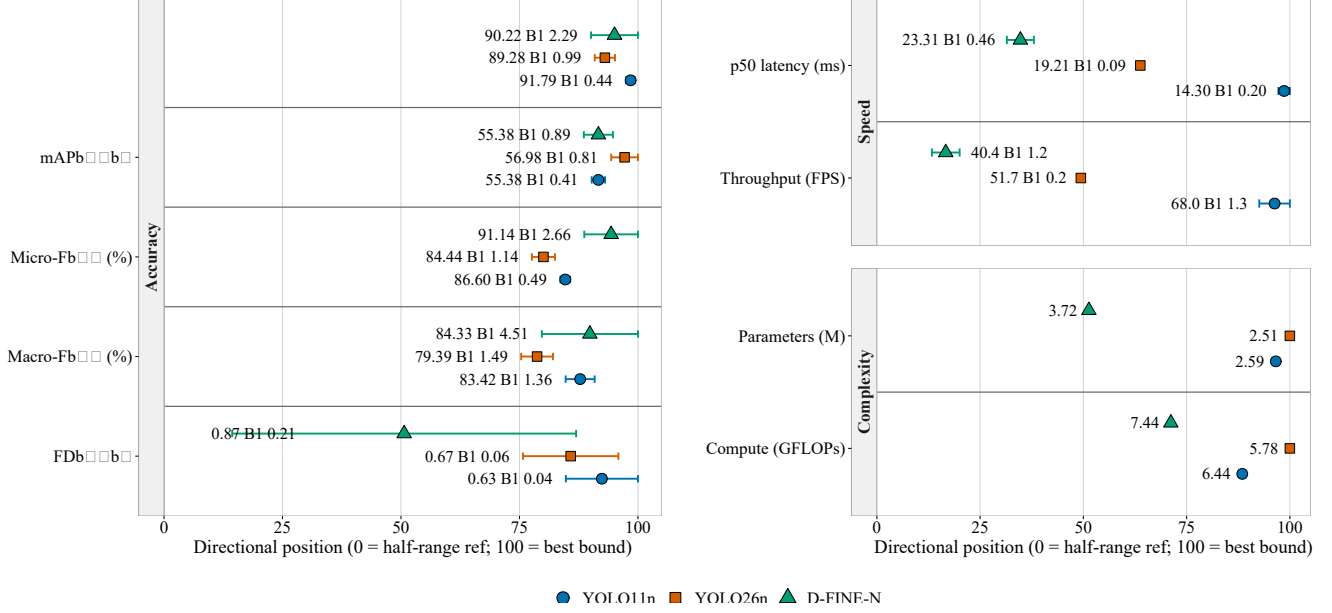


Fig. 2. Directional dot-and-whisker plot for YOLO11n, YOLO26n, and D-FINE-N. Rows are grouped into accuracy (left), speed, and complexity (right); right indicates better performance, and whiskers show sample SD across seeds.

representations, validation-selected thresholds, and the pooled micro- F_1 . It does not directly dominate COCO mAP, because AP is first computed by category and then averaged equally across the four categories.

The class-wise mean differences, YOLO26n minus YOLO11n, are +6.21 points for hand over mouth, -3.85 for yawning, +1.02 for drinking, and +3.02 for phone use. D-FINE-N obtains the highest mean AP for yawning, drinking, and phone use, but the lowest for hand over mouth. Reporting macro- F_1 and per-class AP beside micro- F_1 prevents the majority phone-use class from silently defining the conclusion, although the wide SD for low-support cues still warrants caution.

C. Held-Out Subject Sensitivity

Averaged across seeds, YOLO11n micro- F_1 is 94.94%, 62.41%, and 94.72% for Subjects 05, 10, and 12; YOLO26n obtains 96.11%, 54.11%, and 92.29%; and D-FINE-N obtains 98.20%, 78.79%, and 94.30%. The between-subject change is far larger than the pooled seed SD. Reporting only the pooled test set would therefore hide the difficult driver.

D. Qualitative Error Analysis

Figure 3 shows predeclared seed-13 examples from frozen predictions for all three RGB detectors. Correct detections

localize the intended interaction, while the false-negative examples expose glare and small drinking targets. More importantly, nominal negatives can appear visually cue-like. They may reflect difficult ontology boundaries or missed labels, so the structural annotation audit alone cannot establish semantic validity. The examples are retained because they expose the benchmark’s principal annotation risk.

E. Exploratory NIR Results

Table IV and Fig. 4 reserve the final ten protected-test results. Both artifacts are deliberately marked pending; no value is imputed from RGB or validation. The final analysis will compare each model only between its 1:2 and 1:6 training-negative exposures using the same deterministic midpoint frame for every condition.

V. DISCUSSION

A. Accuracy–Efficiency Trade-offs

No detector is universally best: YOLO11n leads coarse-IoU mAP and throughput, YOLO26n leads strict-IoU mAP and compactness, and D-FINE-N leads micro- and macro- F_1 but runs slowest. Native training and postprocessing make this a shipped-system comparison, not an isolation of architecture.

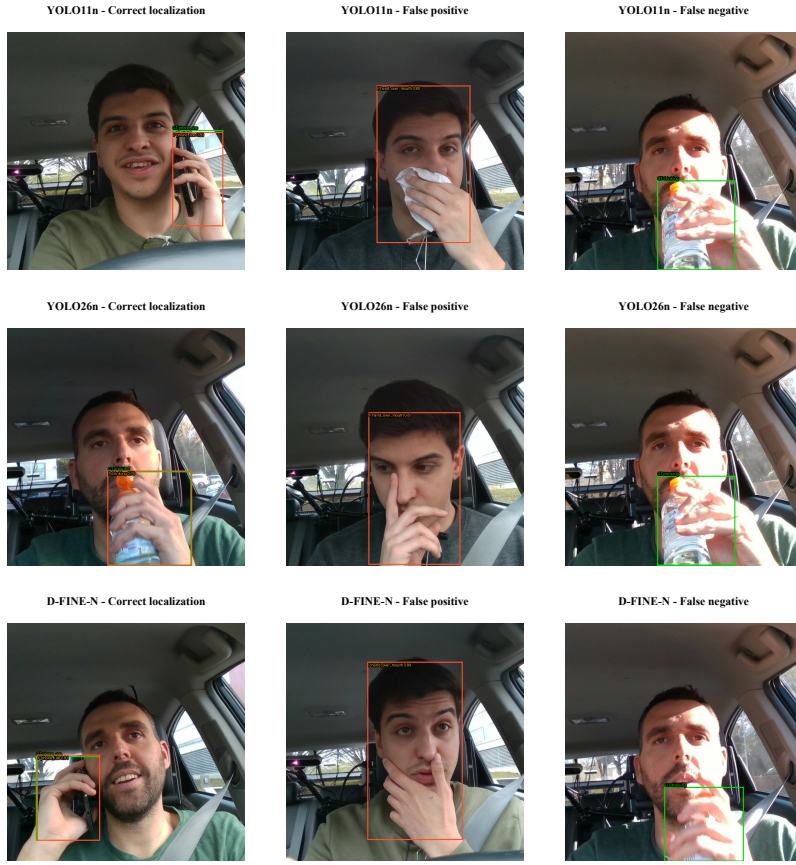


Fig. 3. Representative seed-13 outcomes for YOLO11n, YOLO26n, and D-FINE-N. Green boxes denote ground truth and orange boxes denote predictions at the validation-selected threshold; columns show a correct localization, a false detection on a nominal negative, and a false negative.

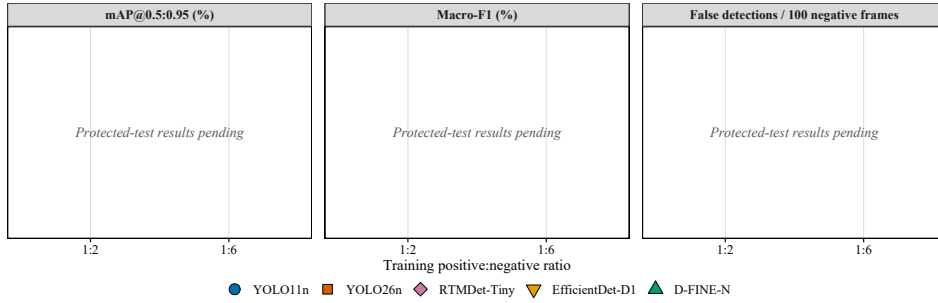


Fig. 4. Pending inventory for ten NIR seed-13 runs at deterministic 1-FPS midpoint sampling. Final panels will show strict-IoU accuracy, macro- F_1 , and false detections per 100 negative frames.

B. Class and Subject Effects

The phone-use majority is a real benchmark characteristic and a limitation for pooled operating metrics, but it does not invalidate YOLO26n's strict-IoU advantage. Equal-weight class AP, macro- F_1 , per-class supports, and paired-seed directions separate that result from majority-class dominance. The narrow aggregate margin, low support for three categories, and drinking sign reversal mean the accurate conclusion is a trade-off rather than a definitive rank. The difficult Subject 10 further shows that seed variation on a single split understates deployment

variation across people.

C. Interpretation Scope

Subject-disjoint evaluation reduces identity leakage but does not establish population-level generalization from only 14 drivers and three held-out test identities. A single annotator and one pass provide internally consistent data but no independent semantic agreement estimate; the qualitative nominal-negative case makes this concrete. Correlated frames remain within RGB recordings, and single-frame detection neither estimates temporal fatigue nor models alert debouncing. NIR avoids

TABLE IV
EXPLORATORY NIR TEST RESULTS AT SEED 13

Model	Neg. mAP _{50:95}	Micro- F_1	Macro- F_1	FD ₁₀₀	p50
YOLO11n	1:2	[PLACEHOLDER]			
YOLO11n	1:6	[PLACEHOLDER]			
YOLO26n	1:2	[PLACEHOLDER]			
YOLO26n	1:6	[PLACEHOLDER]			
RTMDet-Tiny	1:2	[PLACEHOLDER]			
RTMDet-Tiny	1:6	[PLACEHOLDER]			
EfficientDet-D1	1:2	[PLACEHOLDER]			
EfficientDet-D1	1:6	[PLACEHOLDER]			
D-FINE-N	1:2	[PLACEHOLDER]			
D-FINE-N	1:6	[PLACEHOLDER]			

within-snippet duplication through one midpoint frame, but 1-FPS sampling does not characterize motion within the second. NIR additionally differs in dataset, ontology, annotation source, and preprocessing, uses one seed, and changes negative diversity together with training steps. It cannot be interpreted as a controlled RGB-versus-NIR modality effect. Finally, RTX 4060 timing excludes preprocessing and is not an embedded-hardware measurement. These boundaries define what the benchmark supports without removing its value as a transparent, reproducible comparison.

VI. CONCLUSION AND FUTURE WORK

The RGB comparison establishes a system-level trade-off: YOLO11n leads coarse- IoU accuracy and throughput, YOLO26n leads strict- IoU AP and compactness, and D-FINE-N leads pooled and class-balanced F_1 but is slowest. Class, subject, and suite-extension evidence prevent a universal winner or jointly frozen nine-run claim. NIR findings remain [PLACEHOLDER].

Future work will add independent multi-annotator review, increase rare-cue and held-out-driver coverage, repeat the benchmark across subject partitions and external cabins, and test class-balanced or matched-step training. Temporal alert logic will connect frame localization to user-facing warnings. The NIR study will be strengthened with matched optimization exposure and additional seeds, while embedded-device profiling will include preprocessing and end-to-end input latency. These extensions directly target the uncertainty exposed by the present class, subject, and qualitative analyses.

Data and code: Authored annotations, frozen partitions and checksums, sanitized predictions, evaluation tools, and publication artifacts are available at <https://github.com/IamOumarIbrahim/lightweight-driver-behavior-detection>; licensed DMD and Drive&Act media and trained checkpoints remain local under their original access terms.

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